

CARDIOVASCULAR EMERGENCIES

CONSIDER CARDIAC, RESPIRATORY OR GASTROINTESTINAL AETIOLOGY.
CAUSES OF SHOCK = VOLUME, CARDIAC, DISTRIBUTIVE or OBSTRUCTIVE.

PRIMARY SURVEY AND OBSERVATIONS

CORRECT ANY 'ABC' ISSUES CONSIDERING CAUSES OF SHOCK (SEE BELOW)
PERFORM EARLY 12-LEAD ECG, REPEAT IN 15 MINUTES FOR ANY CHANGES

CARDIOGENIC OR OBSTRUCTIVE SHOCK

ACS, P.E., PERICARDITIS INDUCED TAMPONADE, ASTHMATIC INDUCED TENSION PNEUMOTHORAX

CONSIDER ANTIPLATELETS, THROMBOLYSIS, or DECOMPRESSION FOR TENSION

HYPOVOLAEMIC OR DISTRIBUTIVE SHOCK

HAEMORRHAGE, SEVERE DEHYDRATION, BURNS, SEPSIS, ANAPHYLAXIS, NEUROGENIC

CONSIDER VOLUME RESUSCITATION AND VASOPRESSORS (DISTRUBITIVE)

ACUTE CORONARY SYNDROMES (ACS) - CARDIOGENIC SHOCK

PRESENTATION

CENTRAL CHEST PAIN (TIGHTNESS or HEAVINESS), RADIATING TO JAW AND LEFT ARM, DYSPNOEA, ASHEN COLOUR or OTHER SIGNS OF SHOCK, CLAMMY WITH ASSOCIATED RISK FACTORS (SEE BELOW)

ECG

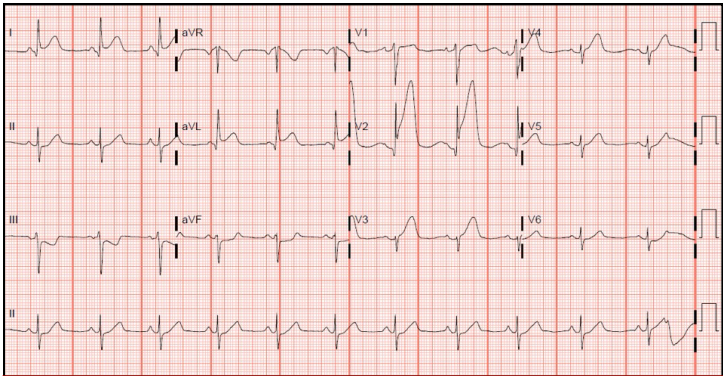
ST ELEVATION (STEMI), ST DEPRESSION, HYPERACUTE T WAVES, WELLENS SIGN, DE WINTER T WAVES (PICTURED BELOW RIGHT).

RISK FACTORS

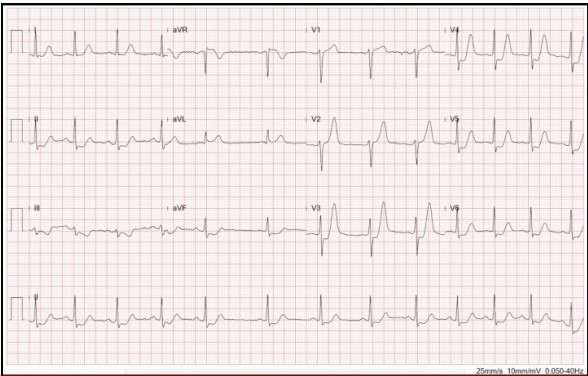
ISCHAEMIC HEART DISEASE, PREV HEART ATTACK, MALE, OBESE, INACTIVE LIFESTYLE, HIGH CHOLESTEROL, SMOKER, FAMILY HISTORY.

MANAGEMENT

MEDEVAC REQUEST, OPIOID ANALGESIA, ANTIPLATELETS (ASPIRIN 300mg), NITRATES (GTN 400mcg), OXYGEN TO MAINTAIN SpO₂ >94%



ANTERIOR STEMI



DE WINTERS T WAVE

AORTIC DISSECTION - HYPOVOLAEMIC SHOCK

PRESENTATION

SUDDEN SEVERE "RIPPING" CHEST PAIN BETWEEN SHOULDER BLADES OR SUBSTERNALLY, APPEARS SHOCKED, DIFFERING RADIAL PULSE PRESSURES or UPPER LIMB BP DIFFERENCE >20mmHg, PULSE DEFICITS, DIASTOLIC MURMUR.

RISK FACTORS

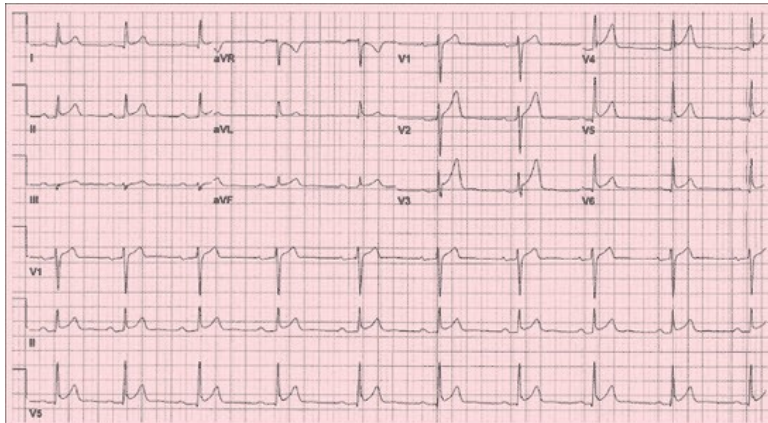
MARFAN or EHLERS-DANLOS SYNDROME, CHRONIC HYPERTENSION, AORTIC DISEASES, HEAVY SMOKER, FAMILY HISTORY OF AORTIC DISSECTION

MANAGEMENT

IMMEDIATE MEDEVAC TO SURGICAL CAPABILITY, PREPARE FOR RAPID DETERIORATION AND RESUSCITATION, GAIN IV ACCESS & GIVE ANALGESIA (TITRATE OPIOIDS), FLUIDS ONLY IF HYPOTENSIVE OR REDUCED MENTATION

PERICARDITIS - OBSTRUCTIVE SHOCK OCCURS DUE TO TAMPONADE

PRESENTATION	CENTRAL CHEST PAIN (SHARP OR PLEURITIC), PERICARDIAL FRICITON RUB, PAIN WORSENS WHEN SUPINE OR ON INSPIRATION ECG = WIDESPREAD ST ELEVATION (ANY DEPRESSION TREAT AS ACS)
CAUSES	RECENT UPPER RESPIRATORY TRACT OR GASTROINTESTINAL INFECTION
RISK FACTORS	MALE, AGED 20-50, PREVIOUS INFECTION, SYSTEMIC AUTOIMMUNE DISEASE
RED FLAGS	SIGNS OF CARDIAC TAMPONADE (SHOCK, JVD, MUFFLED HEART SOUNDS)
MANAGEMENT	NSAIDS (ASPIRIN or IBUPROFEN), IF NO RESPONSE TO NSAIDS IN 7 DAYS THEN REQUIRES MEDEVAC. ANY RED FLAGS REQUIRES IMMEDIATE MEDEVAC



**WIDESPREAD ST ELEVATION FOR PERICARDITIS.
IF THERE IS ANY ST DEPRESSION, THEN TREAT AS ACS**

PULMONARY EMBOLISM (PE) - OBSTRUCTIVE SHOCK & RESP IMPAIRMENT

PRESENTATION	PERC RULE FAILED, SIGNS OF DVT, DYSPNOEA, CHEST PAIN LOCALISED TO AFFECTED AREA OR SUBSTERNAL FOR SADDLE PE, HYPOXAEMIA, TACHYCARDIA, RV STRAIN SIGNS ON ECG, WELLS SCORE >4.
RISK FACTORS	DVT, LONG HAUL FLIGHT, PREVIOUS SURGERY, INACTIVE LIFESTYLE, RECENT FRACTURE (CONSIDER FATTY EMBOLISM), PREGNANCY, BED BOUND >5 DAYS
RED FLAGS	HAEMODYNAMICALLY UNSTABLE (LOW BP, HIGH HR, NO PALPABLE RADIAL PULSE) SEVERE DYSPNOEA (RR >30), SpO ₂ <90%, REDUCED CONSCIOUSNESS
MANAGEMENT	OXYGEN FOR HYPOXIA, ANALGESIA FOR PAIN. REACHBACK & CONSIDER THROMBOLYTIC AGENT IN THE CARDIAC ARREST PATIENT WITH POSSIBLE PE.

WELLS SCORE

SCORE OF <2 CORRESPONDS TO A 1.3% RISK OF PE. SCORE OF ≤4 RESULTS IN A RISK OF 3-12.1%. SCORE >4 HAS A 37% RISK OF PE. USE CLINICAL JUDGEMENT AND USE REACHBACK IF REQUIRED.

WELLS ≤ 4 = LOW RISK "PE UNLIKELY"

WELLS >4 = HIGH RISK "PE LIKELY" >37%

CLINICAL SIGNS & SYMPTOMS OF DEEP VEIN THROMBOSIS (DVT)	3 POINTS
PE IS PRIMARY OR EQUALLY LIKELY DIAGNOSIS	3 POINTS
HEART RATE >100 BEATS PER MINUTES	1.5 POINTS
IMMOBILISED FOR AT LEAST 3 DAYS OR SURGERY IN THE PAST 4 WEEKS	1.5 POINTS
PREVIOUSLY OBJECTIVELY DIAGNOSED WITH DVT OR PE	1.5 POINTS
HAEMOPTYSIS	1 POINTS
MALIGNANCY WITH TREATMENT WITHIN PAST 6 MONTHS OR ON PALLIATIVE CARE	1 POINTS